Puma Marketing Strategies and Parkruns

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Overview

In this study, we analysed 11 weeks of Parkrun events across 7 locations in and around Christchurch/Otautahi. This study was commissioned by Puma Inc. for the purpose of determining marketing potential in these Parkruns, and developing and optimal marketing strategy. This study was conducted by Bad Ass Analytics Ltd. and this document investiagates these Parkruns in relation to the potential for marketing/advertising.

Key Insights

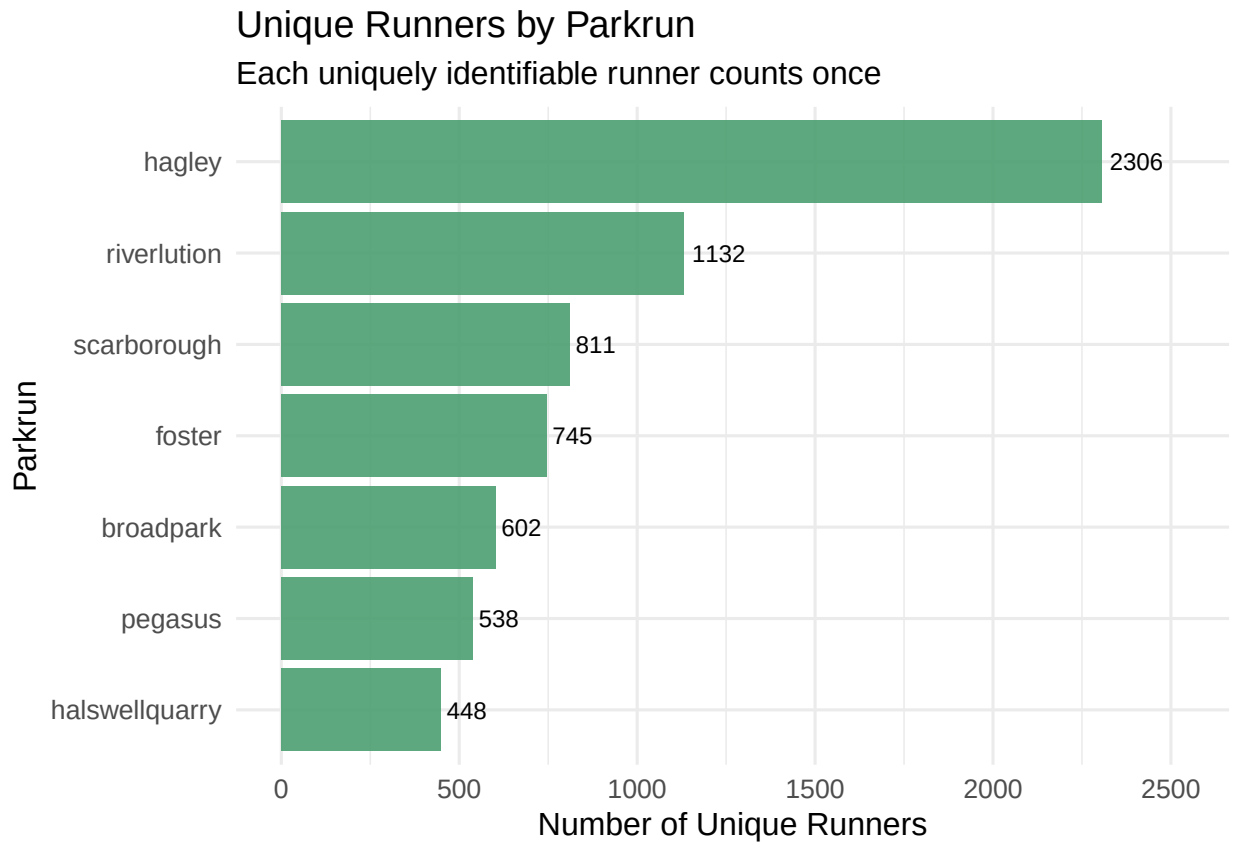
Question One: Which Parkrun provides the best marketing opportunity with the largest number of people?

This question surrounds determining which Parkrun location generally has the most runners, and assumes that all runners will be a potential customer.

Table 1: Unique runners and total finishes per Parkrun

Parkrun	Total Finishes	Unique Runners	Unknown Finishes
hagley	6081	2306	835
riverlution	2055	1132	162
scarborough	1883	811	218
foster	2087	745	74
broadpark	1285	602	67
pegasus	1525	538	29
halswellquarry	805	448	40

Table 1 shows that over the past 11 weeks, the Hagley Parkrun has had the highest number of unique runners out of all Parkrun locations at 2306.



Following on from Table 1, Chart 1, as seen above, also shows that Halgey is the most popular Parkrun, followed by Riverlution, and Scarborough. As such, advertising at these locations would be a very cost effective way of marketing Puma products to a very large audience.

Question Two: Do new people come each week or is it always the same people?

We can determine whether or not if new people come each week, by counting the number of First Timers that have ran in any Parkrun over the past 11 weeks.

Firstly, we have to define First Timer. A First Timer is a runner who has never ran at a particular Parkrun location before attending an event at it. As such, First Timers may have never ran a Parkrun before at all, or may have ran at other locations, but not this one.

Table 2, as seen above, shows the number of First Timers that have ran at every Parkrun over the past 11 weeks. The total number of First Timers who have ran in the past 11 weeks is generally about 20% of all of all runners who ran that week. This generally suggests that most runners are repeats.

Table 2: Weekly new vs returning runners across every Parkrun

Week	First Timers	Total Runners	Percentage of First Timers
1	620	1900	32.63
2	358	1675	21.37
3	225	1303	17.27
4	196	1097	17.87
5	291	1346	21.62
6	250	1134	22.05
7	180	1118	16.10
8	285	1312	21.72
9	205	1187	17.27
10	168	1060	15.85
11	202	1164	17.35

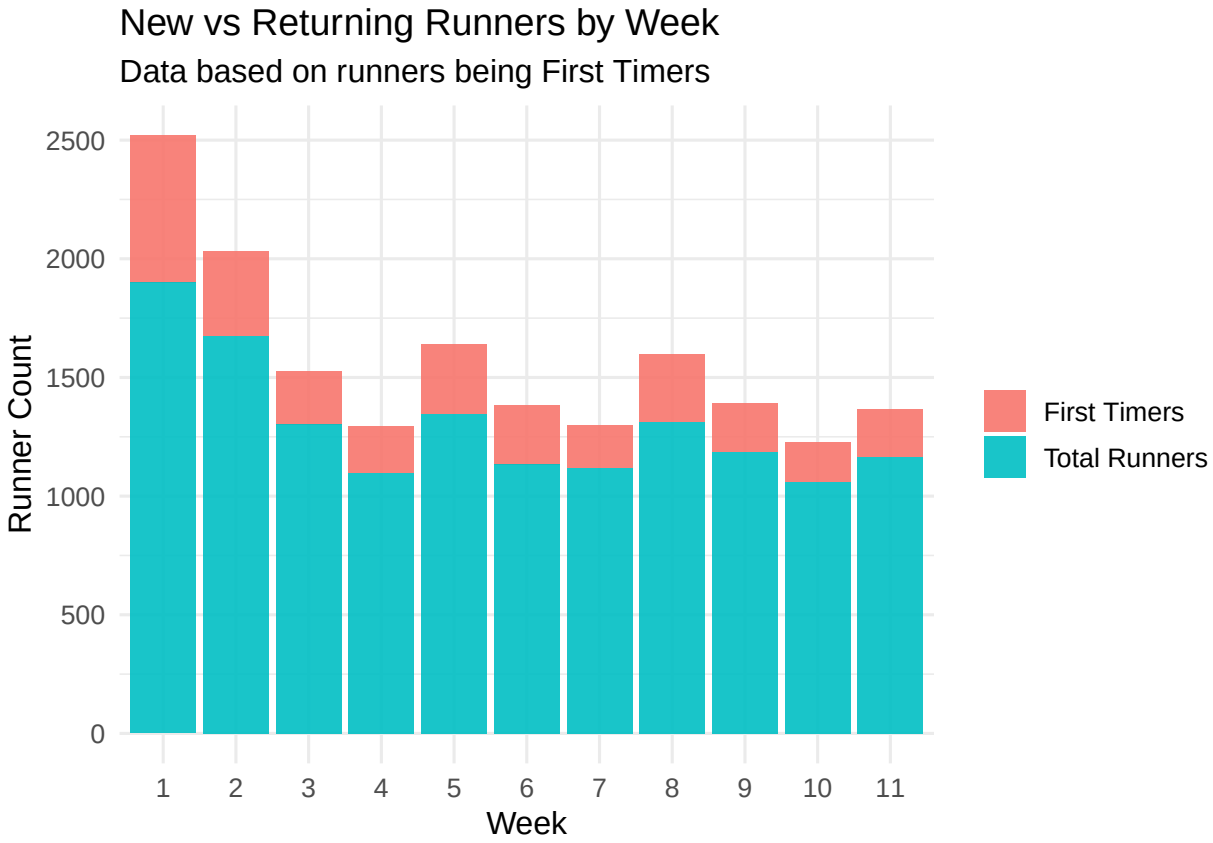


Chart 2 above corroborates this hypothesis, and shows that the proportion of First Timers to Repeat Runners is generally consistent.

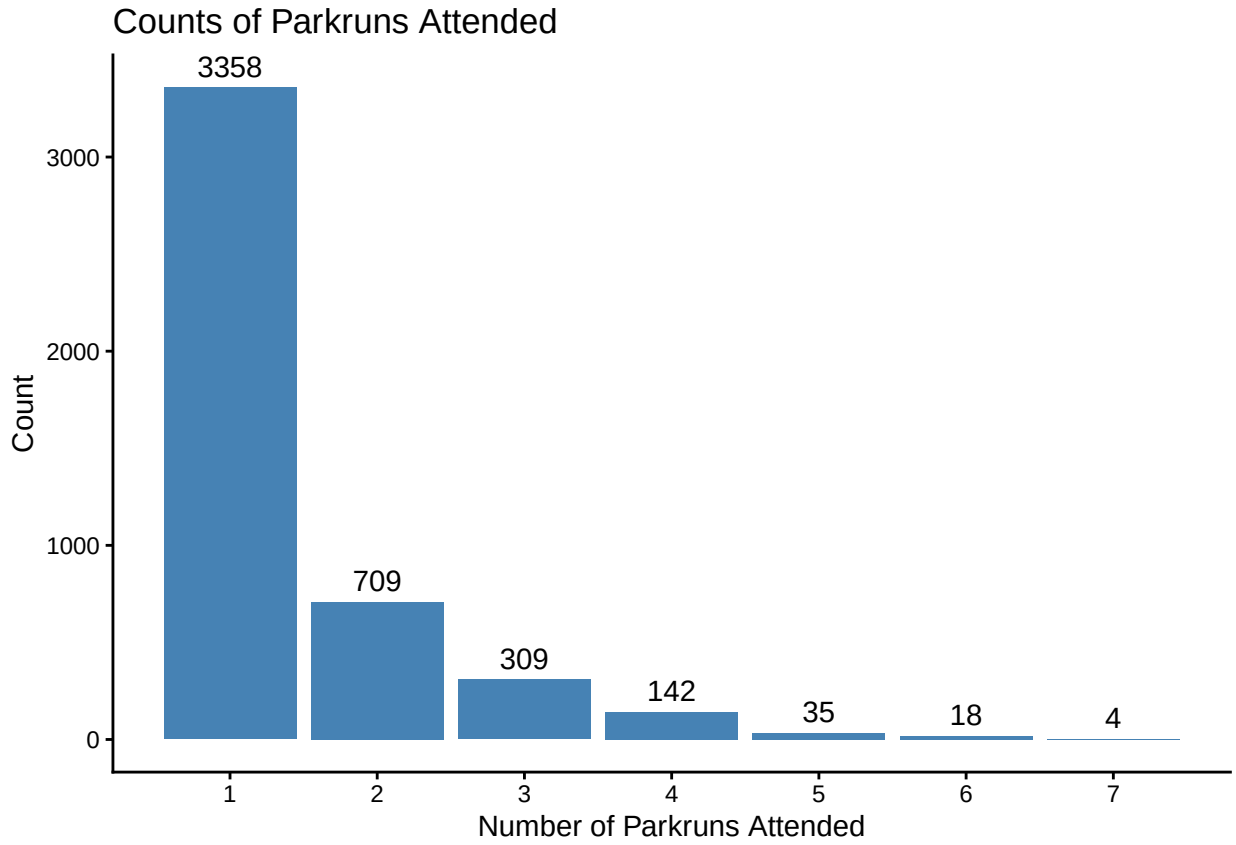
In terms of marketing, this means that there is potential for new customers all of the time, however advertising for prolonged periods of time may not be beneficial, as many runners may already be customers due to advertising at prior events.

Question Three: Do people move between Parkruns? How does this affect advertising?

In order to determine whether or not runners attend different Parkruns, we can simply count the number of Parkrun locations each runner has been to in the past 11 weeks. However, due to unidentifiability, we cannot include any unknown runners in this question.

Table 3: Proportion of runners who have attended a different parkrun

Total Runners	Mixed Runners	Percentage of Mixed Runners
4575	1217	26.6



The above table (Table 4) shows that there is a significant number of runners who have attended Parkruns at mulitple locations (26.6%), or about 1200 runners out of 4600. However, the chart shows that the vast majority of runners have only attended events at one Parkrun location, with very few reaching out to others.

However, this 26% overlap means that advertising across all locations is not going to be particularly budget effective, as advertisements will end up being displayed to people who have already bought Puma products. Therefore, advertising across fewer Parkrun locations can be a more economical option, as it would allow for the same effective advertisement reach, at a lower cost when compared to advertising at all locations.

Question Four: If Puma want to target elite runners, where should they focus their efforts?

In order to target elite runner for the marketing campaign, we can look at the "Age grade" of each runners. "Age grade" is a percentage that compares runner's time to the world record for runner's age and gender.

This percentage could be understood as: - <50% = beginner

- 50-60% = average
- 60-70% = good
- 70-80% = very good
- 80%+ = excellent

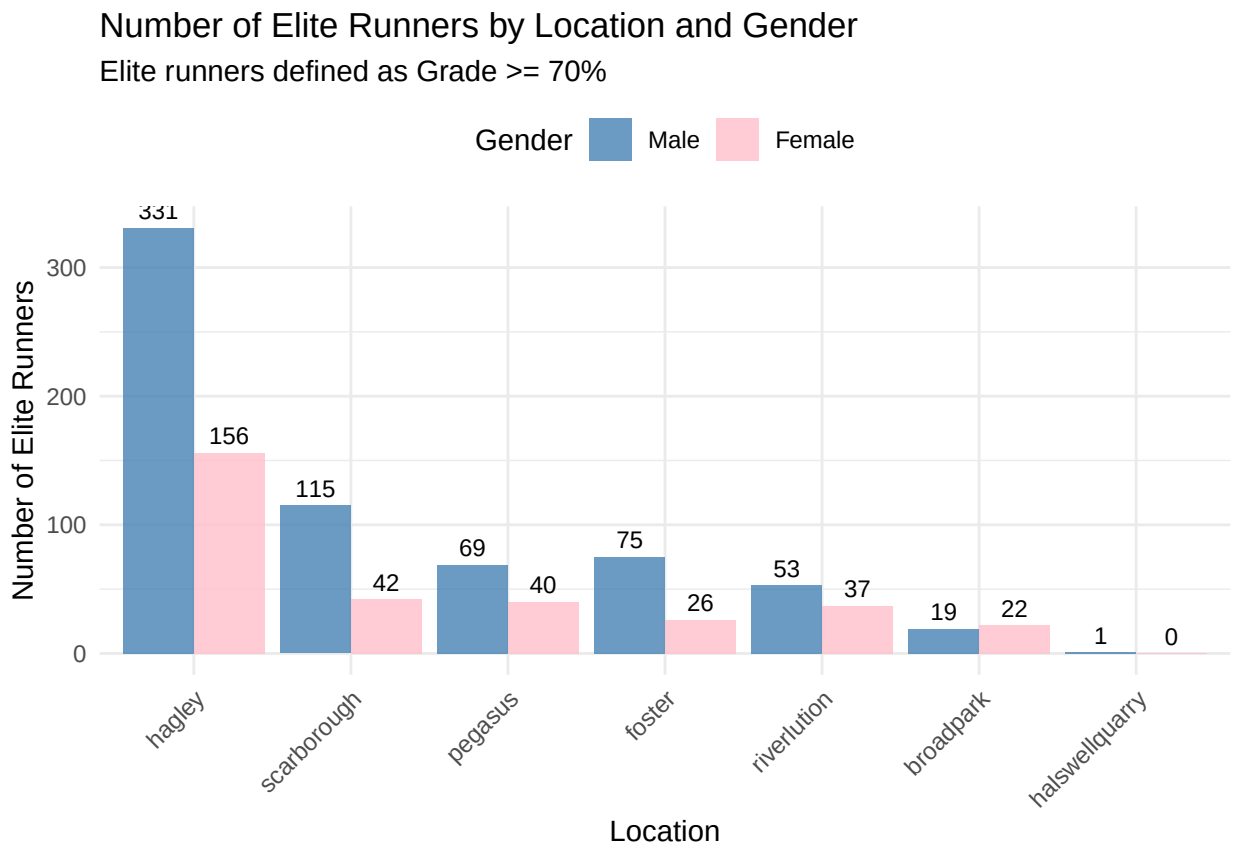
A person who has an age grade above 70 would be consider as an elite runner, this applies to both men and women. Here is the table shows the number of elite runners between parkruns.

Table 4: Overal Elite Analysis by Parkrun Location

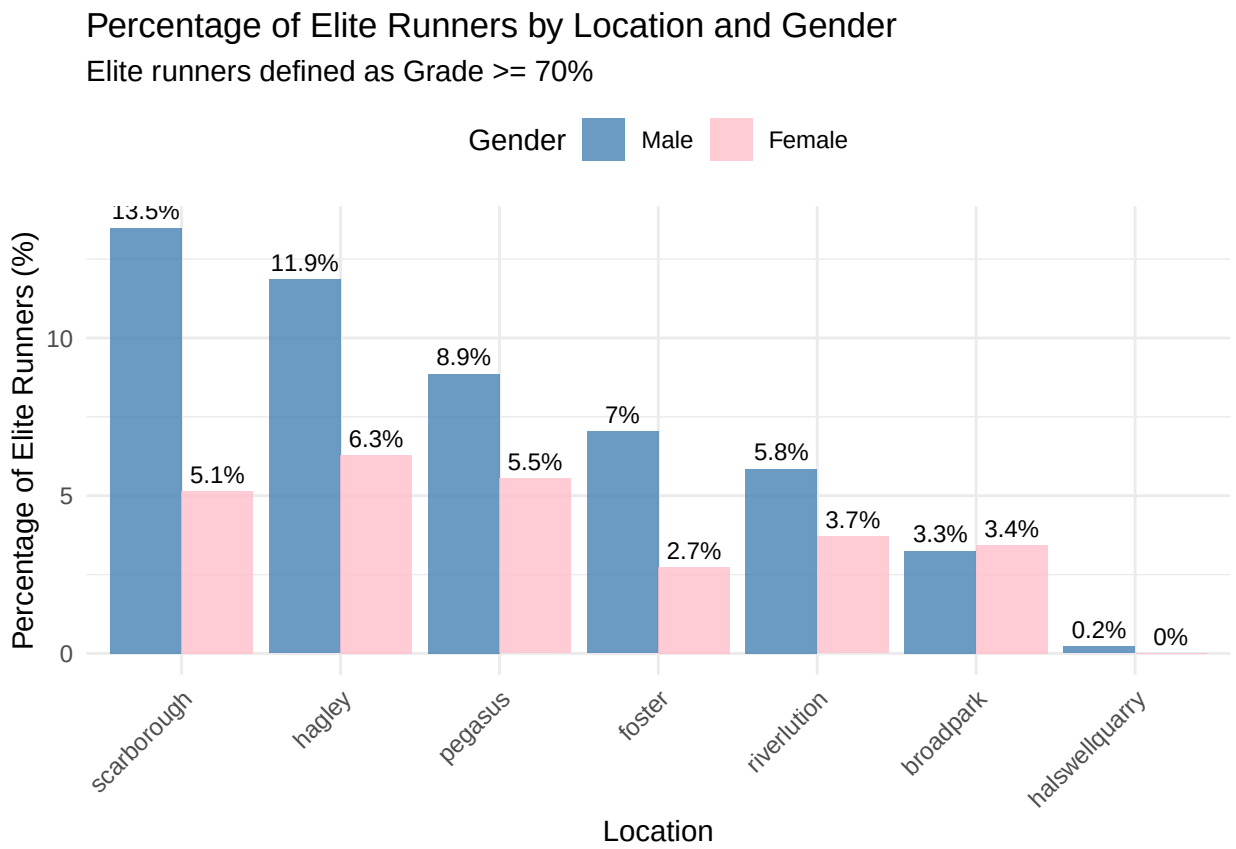
Parkrun Location	Total Runners	Elite runners	Elite runners percentage	Average grade
hagley	5274	487	9.23	53.02
scarborough	1669	157	9.41	53.71
pegasus	1500	109	7.27	52.24
foster	2018	101	5.00	50.45
riverlution	1904	90	4.73	49.88
broadpark	1225	41	3.35	48.81
halswellquarry	769	1	0.13	43.18

It's noticeable to find that the Hagley parkrun has an outstanding numbers of elite runner but Scarborough parkrun has the highest elite runner percentage and highest average grade among these parkruns.

Here is a bar chart to demonstrate the difference number of elite runners by location and gender. The number of men elite runners tend to double than the number of women elite runner in every parkruns. Hagley parkrun has the most both men and women elite runner, second is Scarborough, third is Peagsus.



But if we look at the percentage distribution, Scarborough has the most percentage of men elite runner then Hagley and Pegasus. And the most percentage of women elite runner belongs to Hagley.



To make Puma campaign targeting elite runners effective, Hagley parkrun is the most suitable place. Second place would be Scarborough then Pegasus.

Question Five: What locations feature the most consistent runners?

Another potential customers group is the consistent runners. The table shows the number of consistent runners group by location. Consistent runners would be recognized if they had attended from 3 to 4 weeks, we would consider highly consistent runners if they had attend from more than 5 weeks. And a runner joined 1 or 2 weeks would be considered as occasionally.

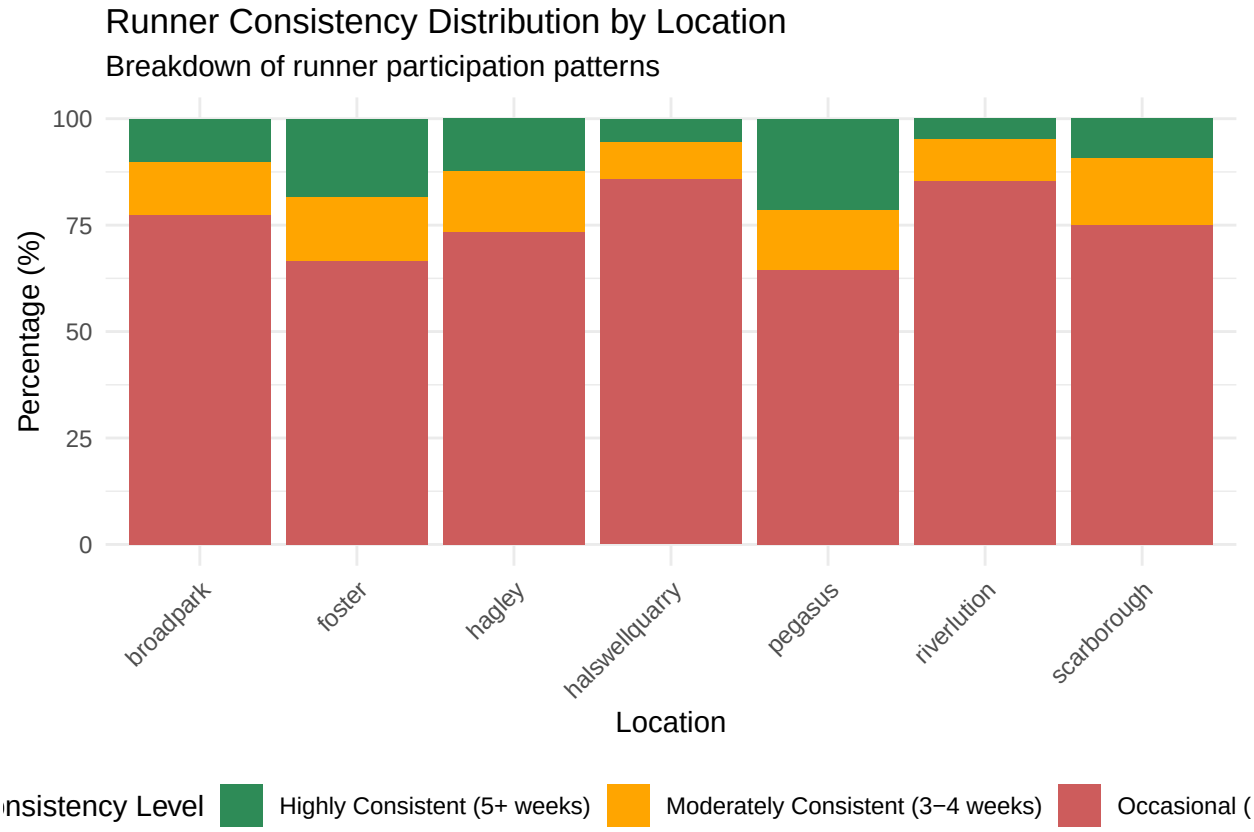
Table 5: Number of consistent runner in each parkrun

Parkrun Location	Total Unique Runners	Highly Consistent (5+ Weeks)	Moderately Consistent (3–4 Weeks)	Occasional (1–2 Weeks)	Percentage Highly Consistent (%)	Average Weeks per Runner
pegasus	539	116	76	347	21.52	2.75
foster	745	138	111	496	18.52	2.66
hagley	2300	285	330	1685	12.39	2.24
scarborough	809	75	128	606	9.27	2.02
broadpark	603	61	75	467	10.12	1.99
halswellquarry	449	25	39	385	5.57	1.70
riverlution	1131	56	110	965	4.95	1.66

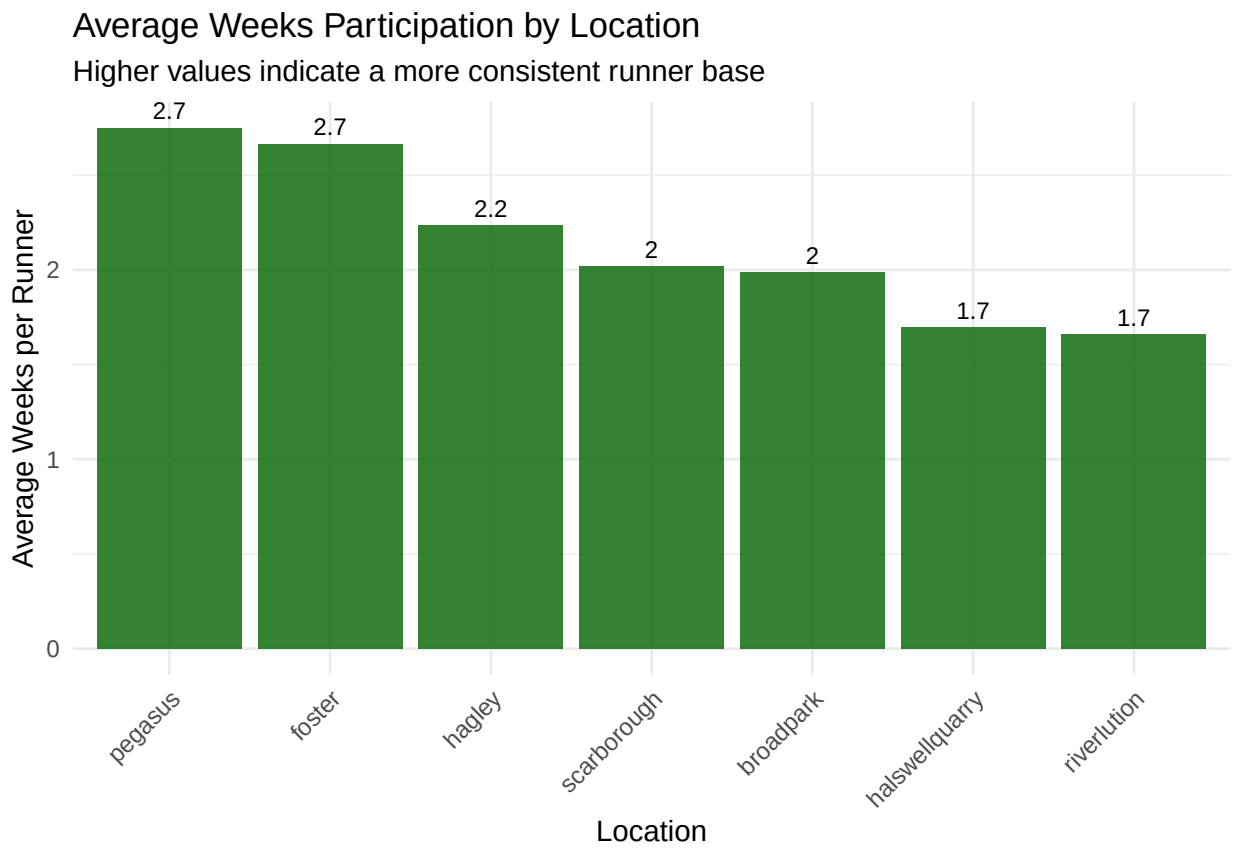
There is no doubt that Hagley parkrun has the most number of highly and moderately consistent runners. Foster and Pegasus parkrun has the second and third of number of highly consistent runners consecutively.

Scarborough only place the fourth in highly consistent runners but it has the second most of the moderately consistent runner. Also the number of moderately consistent runners in Riverlution parkun is quiet noticeable because the dataset for Riverlution is just 8 weeks in stead of 11 weeks like other parkrun but it has almost equal amount as Scarborough and Foster.

In the number aspect, Hagley has the most number of consitent runners but when it come to percentage, Pegasus parkrun has the most percentage of highly consistent runners. At the second and third place are Foster and Hagley consecutively.



And if we compare the average weeks participation among location, it has the same order of top 3.

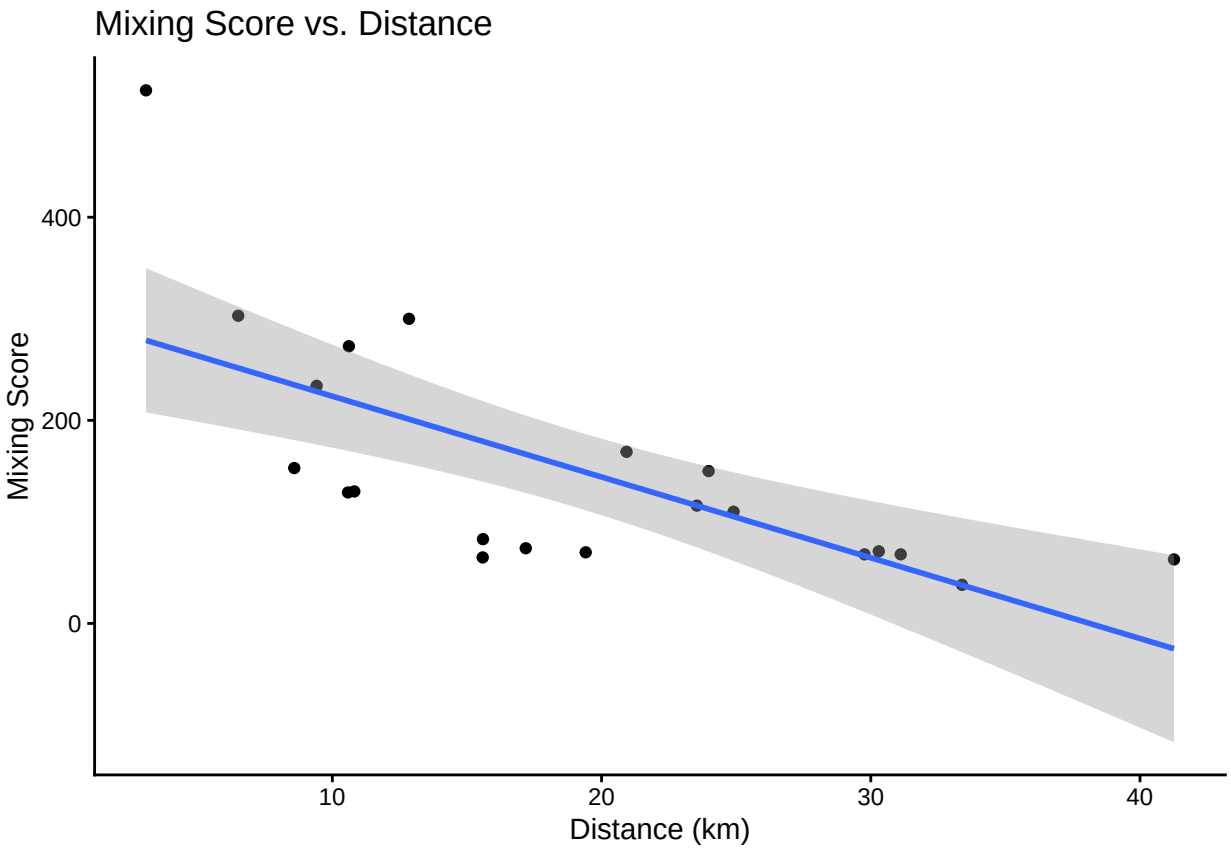


If Puma want to reach out the most number of consistent runners, Hagley will be the best option. Although, the percentage of the consistent runners and the average weeks participated of each runner in Pegasus and Foster are high, the number of highly consistent runners are just a half of Hagley. On the other hand, the number of consistent runners in Riverlution is promising if Puma want to focus on consistent runners as the potential customer.

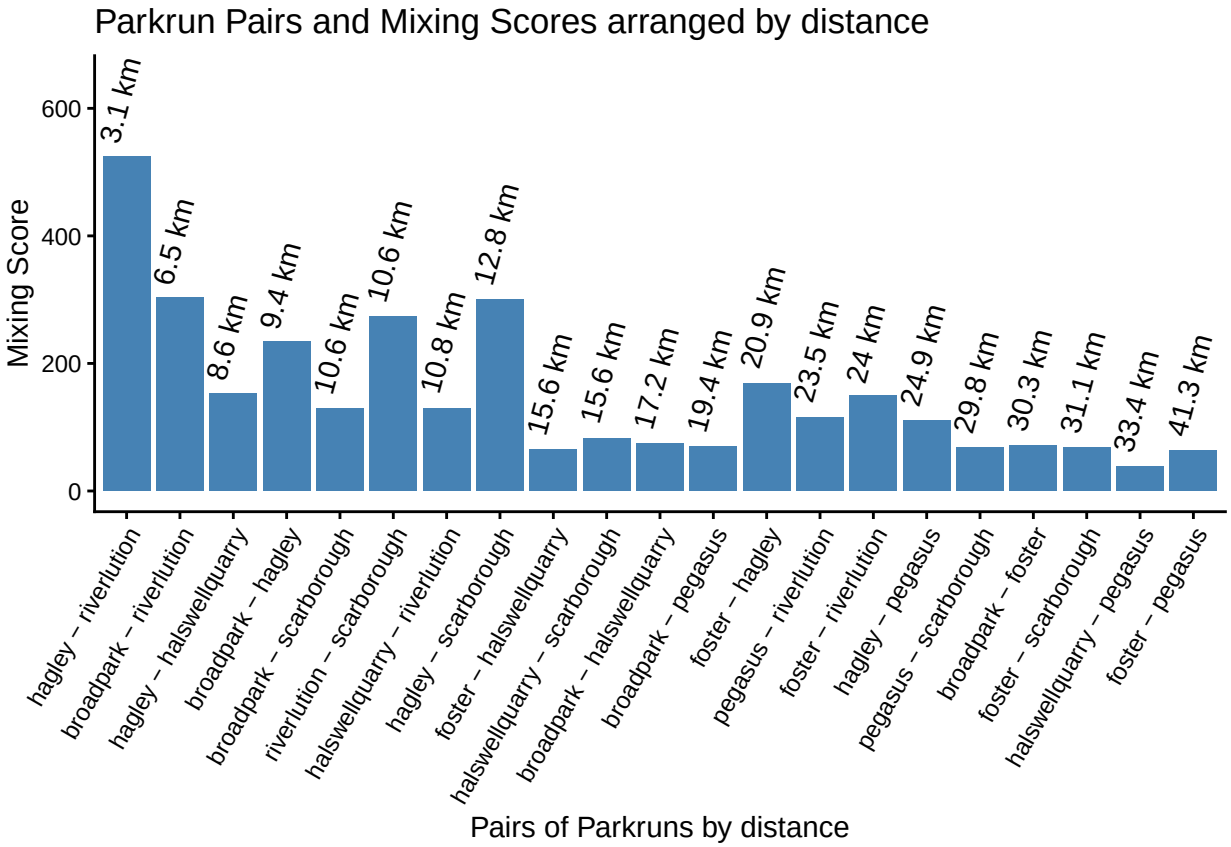
Question Six: Does distance between Parkruns affect how much mixing occurs?

In order to determine whether or not the distance between two Parkruns predicts how much mixing occurs, we can obtain each unique pair of Parkruns, and then using geographical data, determine the distances between these pairs, effectively creating a map of each Parkrun location.

We can then count the number of runners who have gone to both locations in these pairs, and set that as our Mixing Score. A higher Mixing Score means more runners have attended both Parkruns in the past 11 weeks.



As seen above, the figure shows that there is a negative correlation between Distance and Mixing Score. In other words, as the distance between parkruns increases, the amount of mixing decreases.



The above bar chart, also shows this correlation, but also labels each parkrun pair with the distance between them. As is seen in the chart, the Hagley and Riverlution parkruns are very close (only 3km apart), and as such, receive a very large amount of mixing.

As for how this affects advertising, this allows for more control over audience overlap. Given the amount of mixing between the Hagley and Riverlution parkruns, advertising at just one of these parkruns would minimise the amount of overlap and spending, without reducing the effectiveness of the advertisements. However, given the amount of distance between the Foster and Pegasus parkruns, advertising at only one location would only harm any marketing efforts.

As such, for pairs of parkruns with high amounts of mixing, advertising at just one location would be the best option, preferably the parkrun which is generally more popular/has the most runners (advertise at Hagley over Riverlution).

Summary and Conclusion

Pooling all of our knowledge that has been collected together, we can come to the following conclusions about how to maximise the effect of advertising and marketing.

1) Hagley is the most popular Parkrun, and therefore provides the best marketing opportunity to advertise to the largest number of people.

2) A lot of runners are repeat runners, so many may already be customers, and are less likely to buy any Puma products.

3) There is mixing between Parkruns. This means that there is audience overlap, so advertising at every location will have diminishing returns on investment and effort.

4) Hagley is the most suitable place to advertise when considering elite runners.

5) Hagley is also the most suitable place to advertise for consistent runners.

6) The mixing of runners between locations is dependent on distance. Prioritise popular locations, that are far away from each other.

We can see that Hagley is by far the best place to focus marketing efforts for the most diverse range of people, hosting the largest number of consistent, elite, and casual runners. Since Hagley is also so popular, marketing at nearby locations will have reduced impact in comparison to advertising at more distant locations, such as Foster or Pegasus.